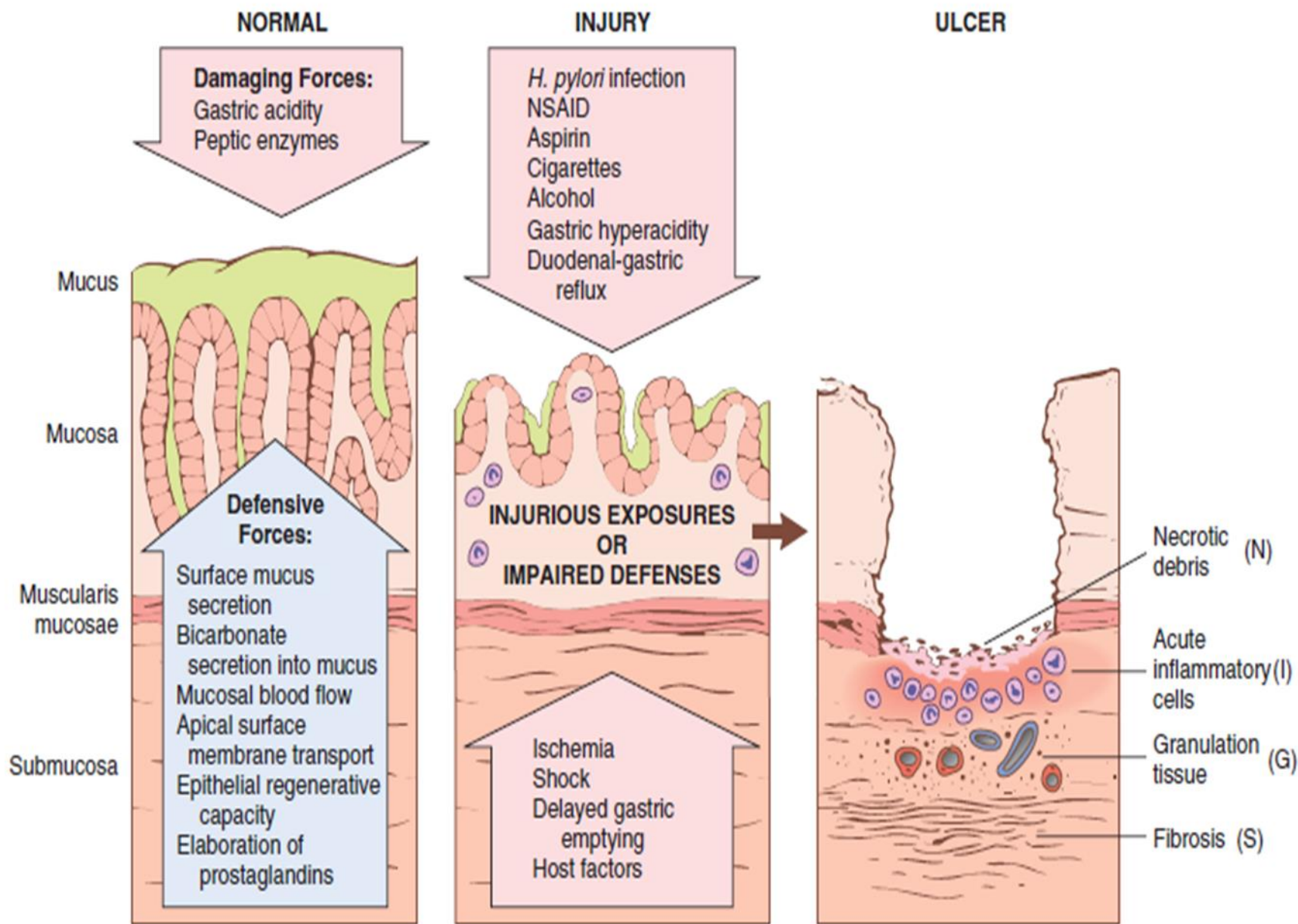




SGL1: Pathology of the stomach

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Mechanisms of gastric injury and protection. This diagram illustrates the progression from more mild forms of injury to ulceration that may occur with acute or chronic gastritis. Ulcers include layers of necrotic debris (N), inflammation (I), and granulation tissue (G); a fibrotic scar (S), which develops over time, is present only in chronic lesions.

Inflammatory diseases of stomach

Acute Gastritis:

- Transient mucosal inflammation.
- May be asymptomatic or cause epigastric pain, nausea, and vomiting.
- Edema & congestion of mucosa, but the surface epithelium is intact.
- In severe cases there may be mucosal erosion, ulceration, hemorrhage, hematemesis, melena, or, rarely, massive blood loss.

Acute gastritis

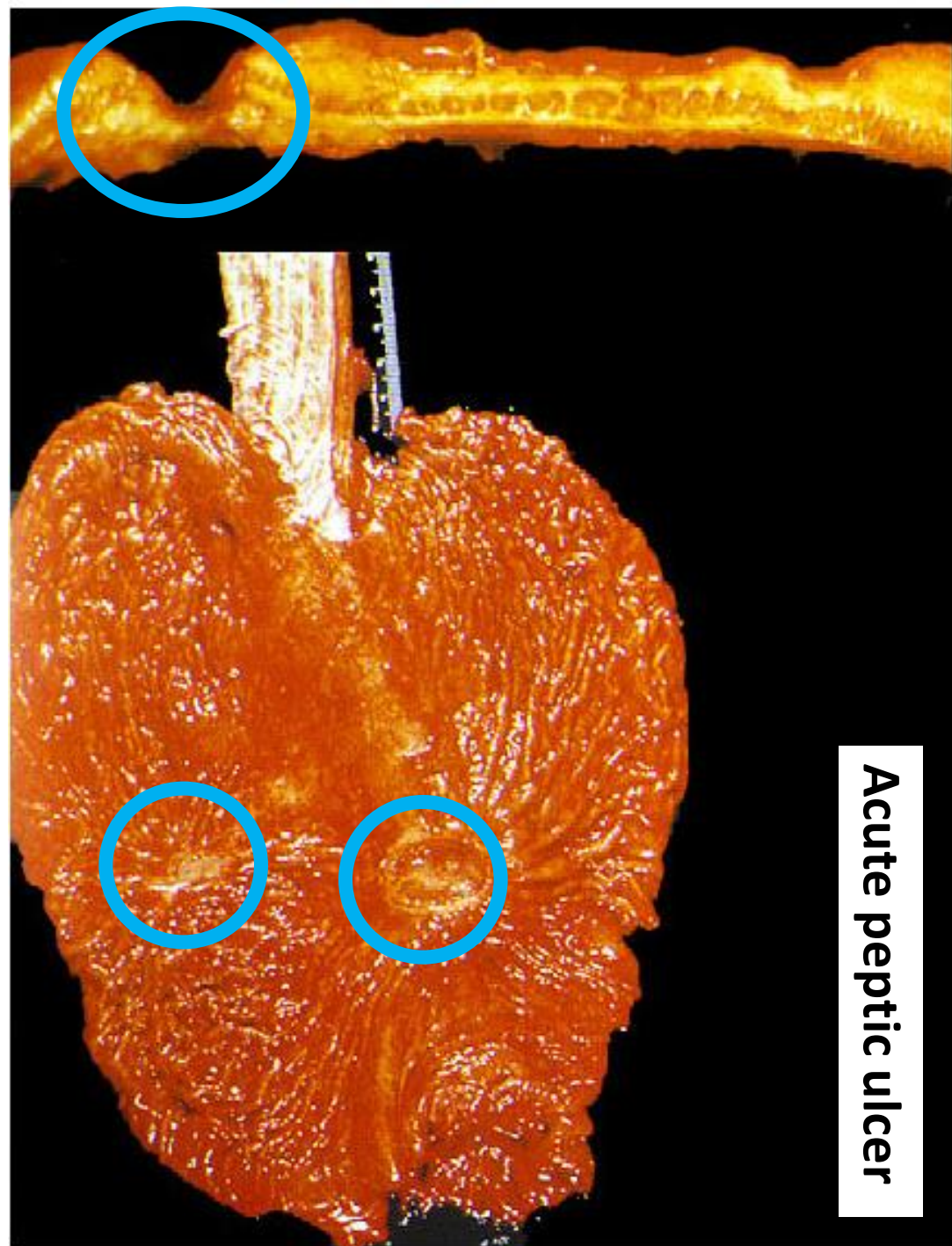


Acute gastritis: There is diffuse and extensive erythema with focal erosions (loss of superficial epithelium above muscularis mucosa).

Acute Peptic Ulceration:

- In ulceration the depth of tissue loss is extend below the muscularis mucosa.
- Causes:
 - Complication of therapy with NSAIDs.
 - Severe physiologic stress which include:-
 - **Stress ulcers:** in patients with shock, sepsis, or severe trauma.
 - **Curling ulcers:** associated with severe burns.
 - **Cushing ulcers:** in persons with intracranial disease.

- There is diffuse hemorrhagic gastritis and **focal peptic ulceration** which is extending nearly through the gastric wall.
- Are rounded and usually < 1 cm in diameter.
- May occur singly or more often multiple.
- Healing with complete re epithelialization occurs days or weeks after the injurious factors are removed.



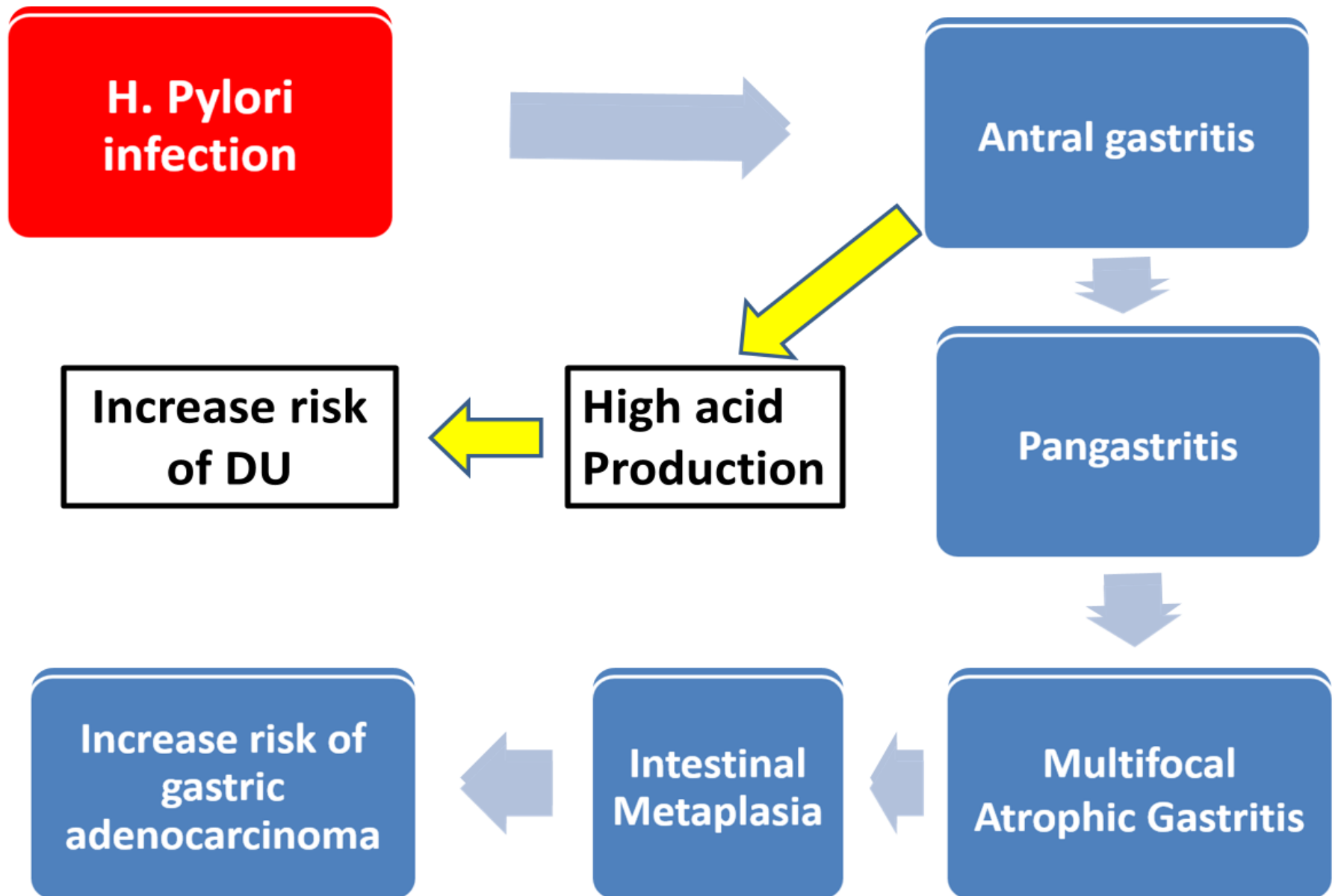
Acute peptic ulcer

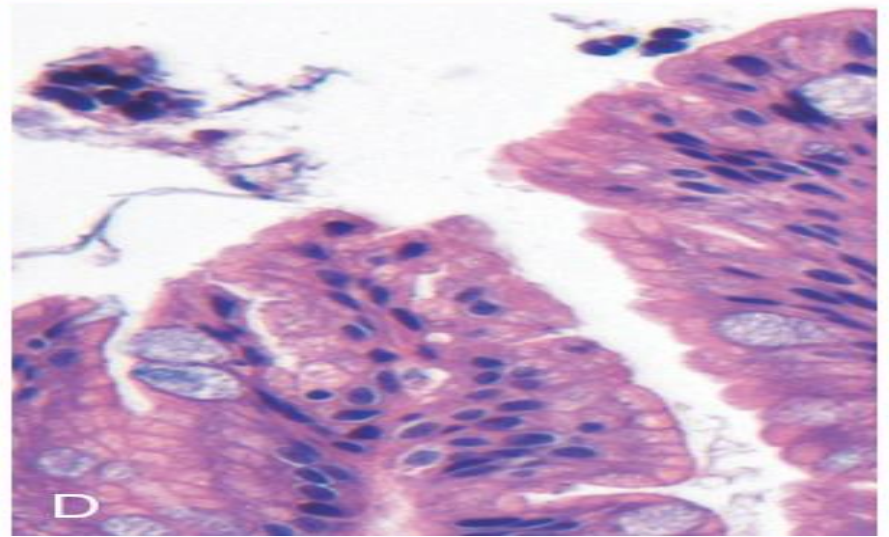
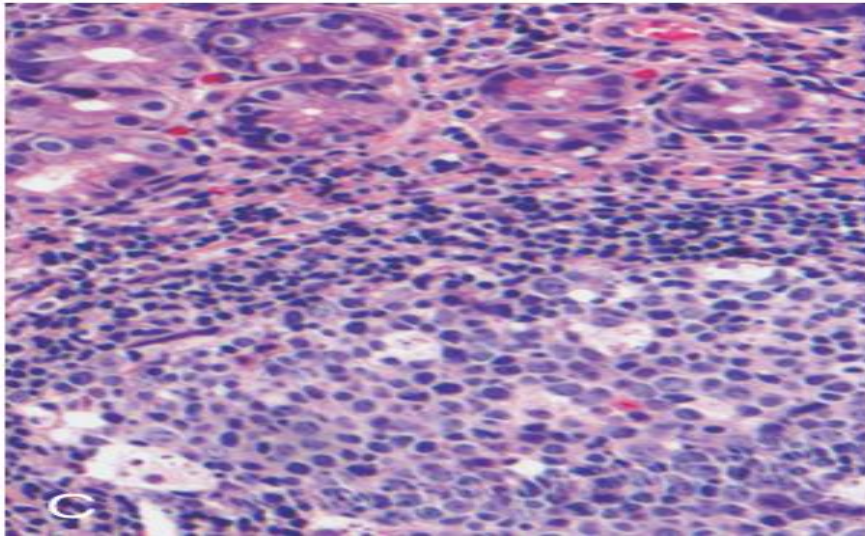
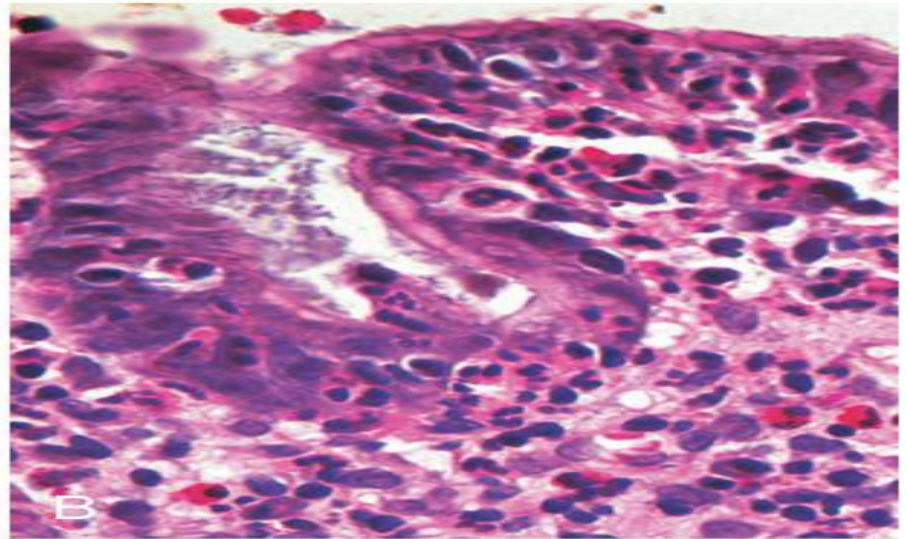
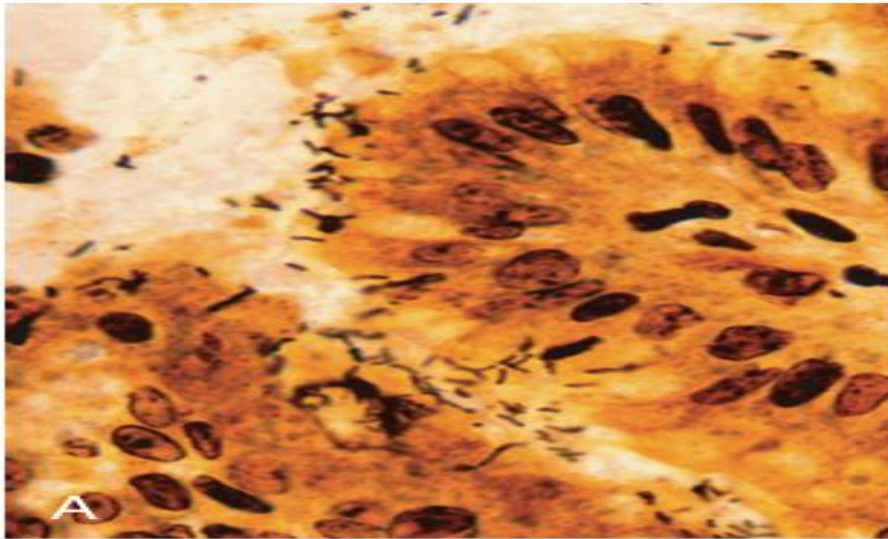
Chronic Gastritis:

- Typically associated with less severe but more persistent symptoms and signs.
- Three major types exist:
 - Helicobacter pylori (H. pylori) induced gastritis: is the most common form.
 - Autoimmune gastritis.
 - Chemical (reactive) gastritis.

H. Pylori infection: may result in:

- Chronic gastritis.
- Peptic ulcer diseases (PUD): H. pylori present in almost all patients with DU, and majority of those with GU.
- Increased risk of gastric adenocarcinoma and gastric MALToma (NHL).



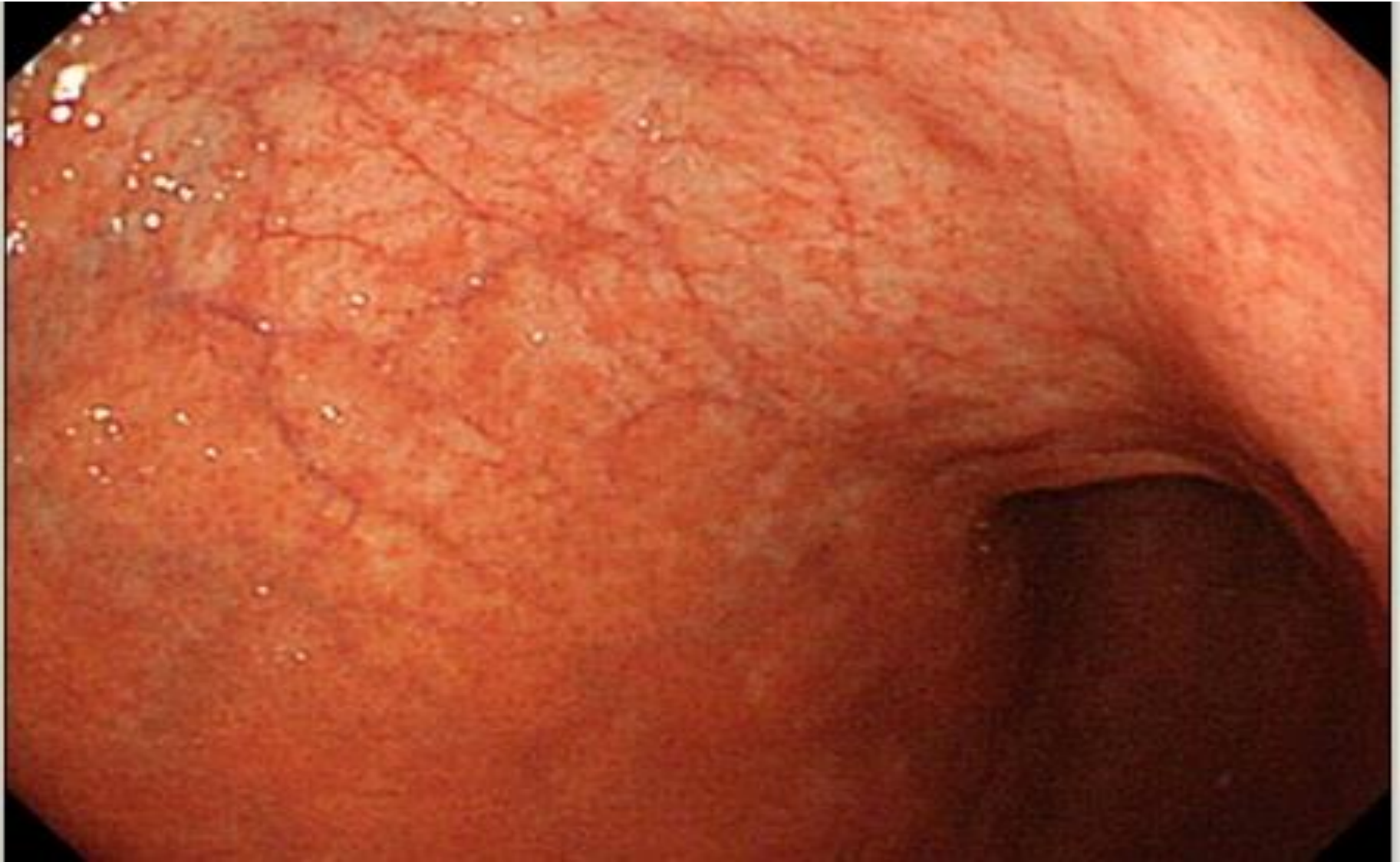


A, Spiral-shaped *Helicobacter pylori* bacilli are highlighted with Warthin-Starry silver stain. **B**, Intraepithelial and lamina propria neutrophils are prominent. **C**, Lymphoid aggregates with germinal centers. **D**, Intestinal metaplasia which is a risk factor for development of gastric adenocarcinoma.

Autoimmune gastritis:

- Represents less than 10% of cases of chronic gastritis, but commonest cause of gastric atrophy.
- Is the most common form of chronic gastritis in patients without H. Pylori infection.
- Pathogenesis:-
 - Autoimmune gastritis is associated with antibodies against parietal cells and damage and loss of these cells which secrete hydrochloric acid (HCL) and intrinsic factor.

- Lack of intrinsic factor disables ileal vitamin B12 absorption, leading to B12 deficiency and megaloblastic anemia (pernicious anemia).
- Although *H. pylori* can cause hypochlorhydria, but it is not associated with achlorhydria or pernicious anemia, because the parietal cell damage is not as severe as in autoimmune gastritis.



Endoscopic appearance of **Gastric atrophy**; Whitish discoloration and visible vessels are evidence of atrophy.

Peptic Ulcer Disease (PUD):

- PUD are **Chronic** lesions that occur in any portion of GIT which exposed to aggressive action of acidic peptic juices.
- PUD may involve the following sites:
 - Stomach: Gastric ulcers (GU).
 - Duodenum: Duodenal ulcers (DU).
 - Lower (Barrett) esophagus.
 - Meckel diverticulum with gastric heterotopia.

- **Pathogenesis:** The imbalances of mucosal defenses and damaging forces that cause chronic gastritis are responsible for PUD (see diagram):-
 - H. pylori infection
 - Drugs: aspirin & NSAIDs
 - Cigarette smoking
 - Increased gastric secretion of HCL acid & pepsin.

- **Morphology:**

- PUD are solitary in more than 80% of patients.
- The classic PUD is a round to oval, sharply punched-out defect.
- The margins are regular with radiating mucosal folds
- The base of the ulcer is smooth & clean as a result of peptic digestion of the exudate.

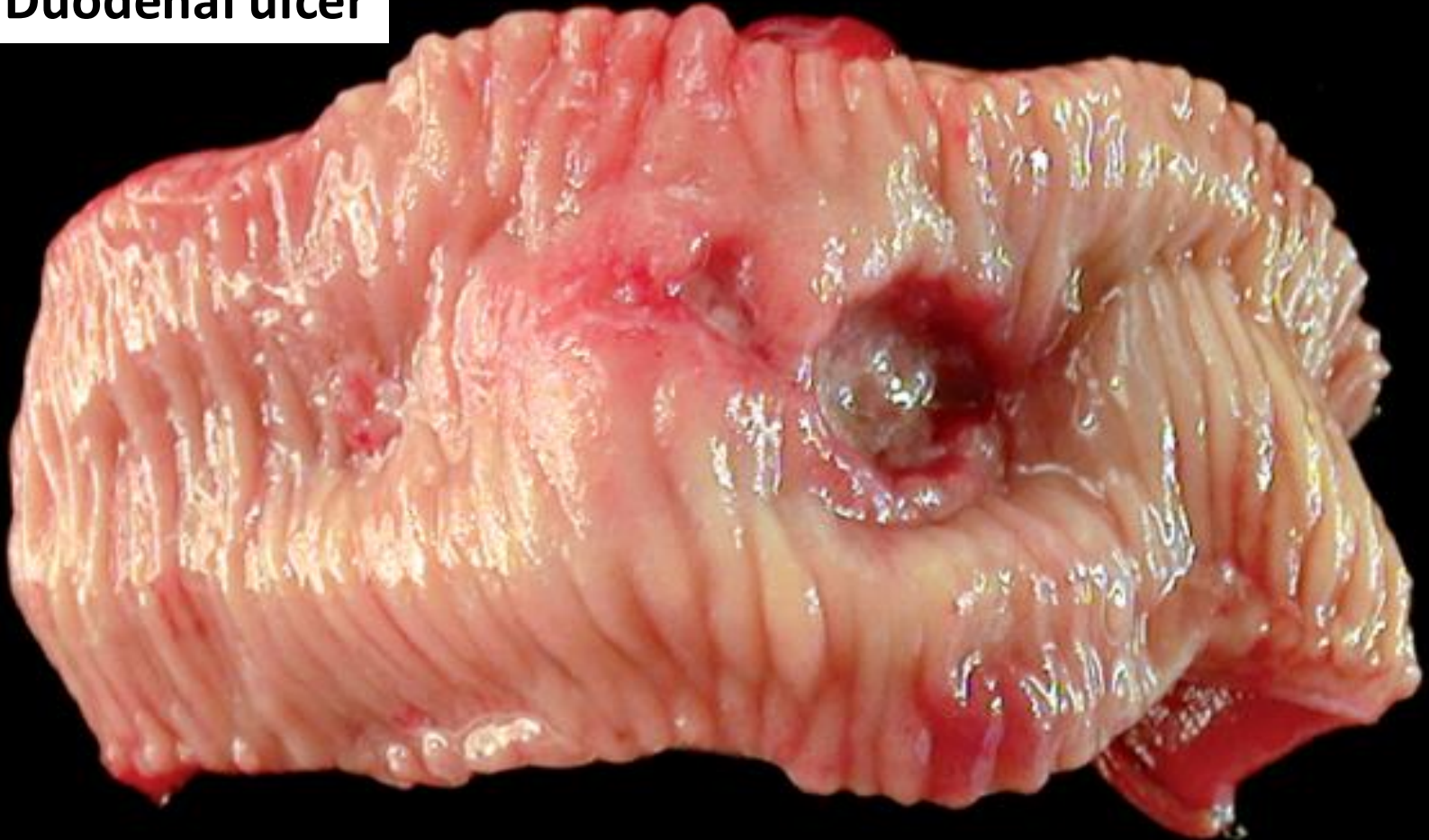
- DU are four times more common than GU.
- DU usually occur within a few centimeters of pyloric valve and involve the anterior duodenal wall.
- GU are usually located near the interface of body and antrum at the lesser curvature site.

Gastric ulcer

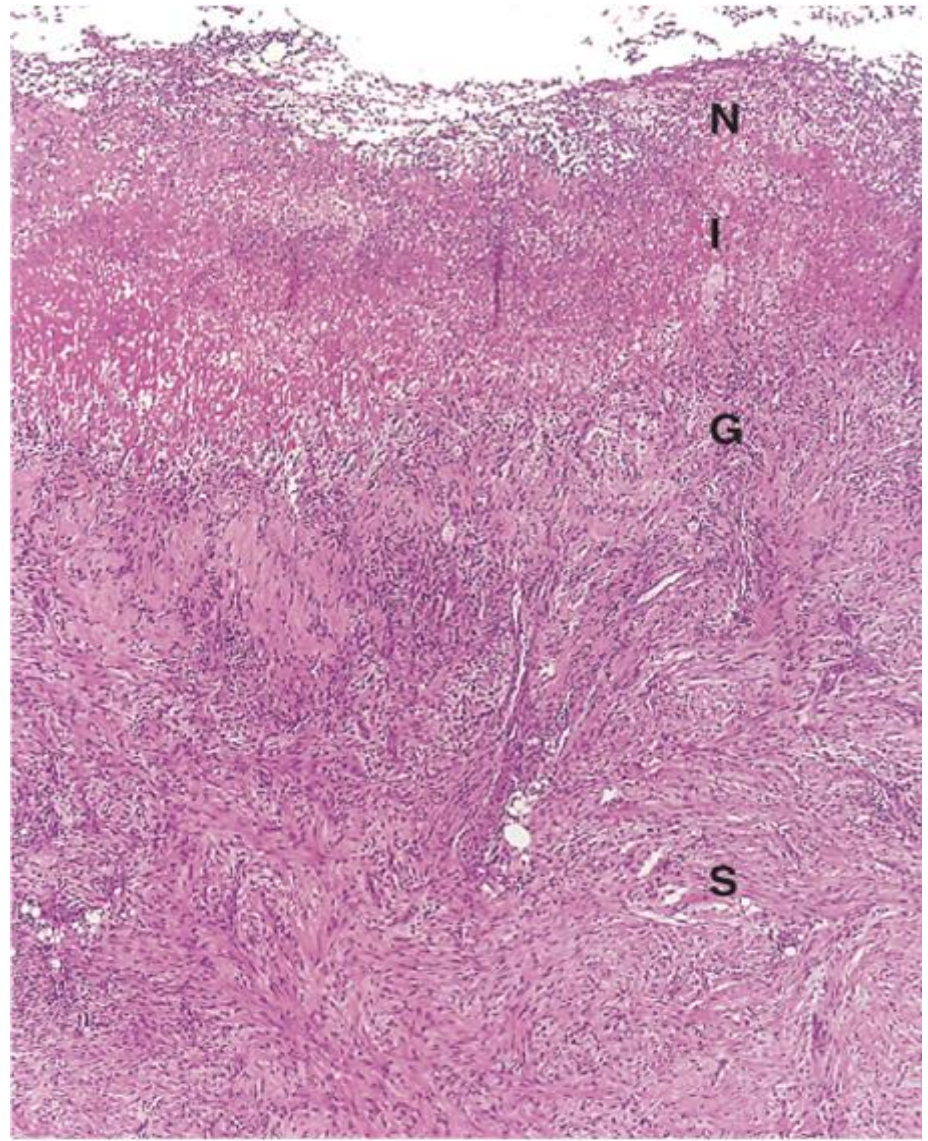
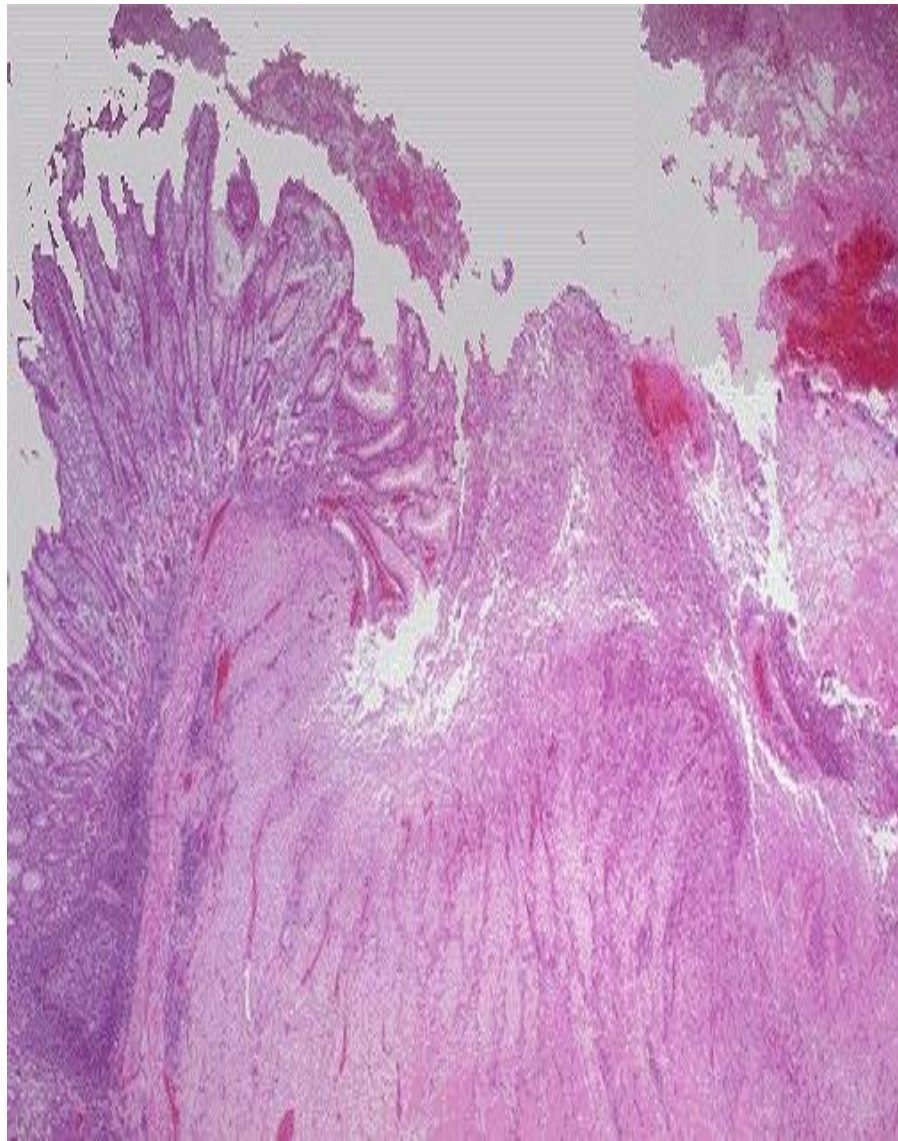


Gastric ulcer: There is a characteristic sharp demarcation from the surrounding mucosa, with radiating gastric folds. The base of the ulcer is gray owing to fibrin deposition.

Duodenal ulcer



Duodenal ulcer: The ulcer is small (2 cm) with a sharply punched-out appearance. Unlike cancerous ulcers, the margins are not elevated. The ulcer base is clean (compare with the ulcerated carcinoma).



PUD share similar histology of the four layers: N= Necrotic area, I = Inflammatory region, G = Granulation tissue, and S = Scar formation (fibrosis).

■ **Complications:**

- Hemorrhage: the most common complication:-
 - Occult.
 - Massive life-threatening.
- Pyloric obstruction (gastric outlet obstruction).
- Perforation and Penetration.
- Development of combined ulcers: GU & DU may occur together.
- Malignant transformation of benign ulcers:-
it is very rare, malignant transformation of GU is more likely than DU.

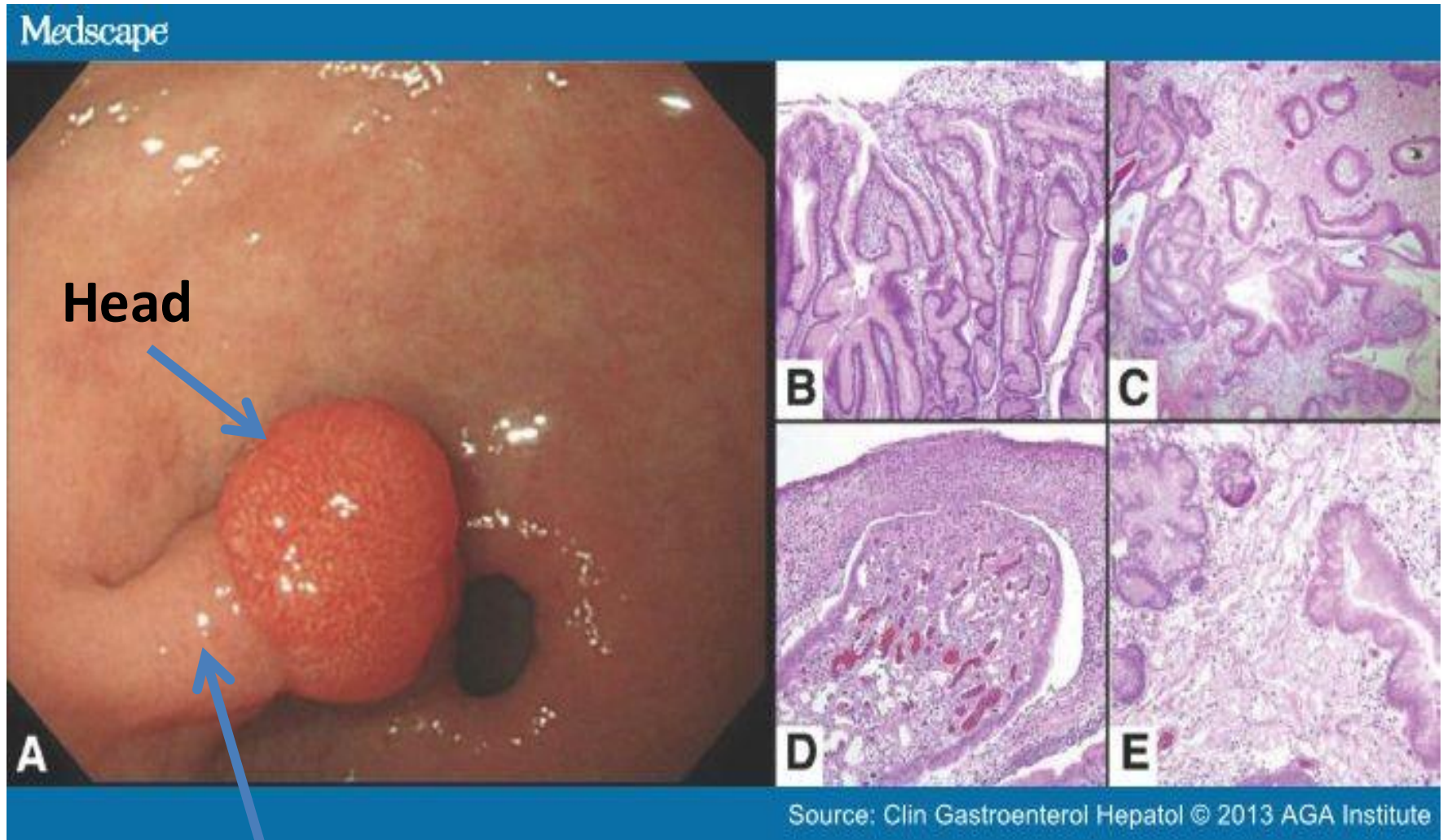
Gastric Polyps and Neoplasms:

Gastric polyps:

- Polyp is any nodule or mass that projects above the level of surrounding mucosa.
- Gastric polyps are uncommon (compared to colonic polyps).

- Types of gastric polyps:-
 - **Hyperplastic / inflammatory polyps:**
the commonest, arise in chronic gastritis.
 - **Fundic gland polyps:**
associated with FAP & chronic use of PPIs.
 - **Adenomatous polyps:**
defined by the presence of dysplasia & have a definite risk of harboring adenocarcinoma.

Hyperplastic polyp: (A) Endoscopic view of a hyperplastic polyp on a stalk in the antrum.



Gastric Cancer:

- Adenocarcinoma is the most common malignancy of stomach (> 90% of all gastric cancers).
- Lymphomas: stomach is the commonest site for extranodal NHL.

Gastric adenocarcinoma:

- Most gastric adenocarcinomas involve the gastric antrum; the lesser curvature is involved more often than the greater curvature.
- The incidence varies markedly with geography.

- Gastric adenocarcinoma rates dropped by more than 85% during the twentieth century:-
 - Possible contributor is the decreased consumption of dietary carcinogens such as N-nitroso compounds and benzo[a] pyrene (because of the reduced use of salt and smoking for food preservation).
 - Mass endoscopic screening programs in regions with high incidence leads to detection of early gastric cancers.

- **Etiologic Factors:** Multifactorial; evident associations are:-
 - Helicobacter pylori: chronic gastritis → gastric atrophy → intestinal metaplasia → epithelial dysplasia → carcinoma.
 - Smoking.
 - Lower socioeconomic groups.
 - Atrophic gastritis, autoimmune gastritis, pernicious anemia.
 - Nitrosamine

- Dietary factors:-

- Increased with: consumption of large amounts of starch, smoked or cured fish, meat, and pickled vegetables.
- Decreased with: consumption of whole milk and fresh vegetables rich in vitamin c and other antioxidants.

- Gastric adenomatous polyps.

- Subtotal gastrectomy.

- GU.

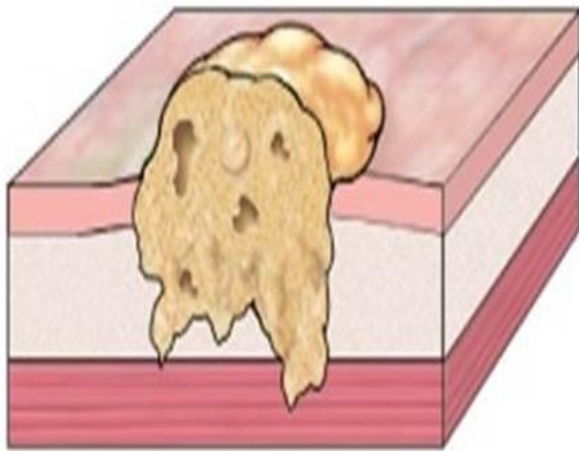
- **Clinical Presentation:**

- Abdominal discomfort.
- Early satiety.
- Nausea and vomiting.
- GI hemorrhage, IDA.

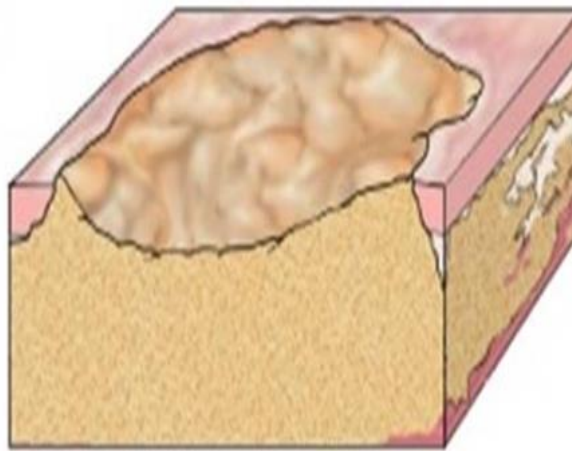
- **Morphology:**

- Gastric adenocarcinoma can be classified grossly as:-

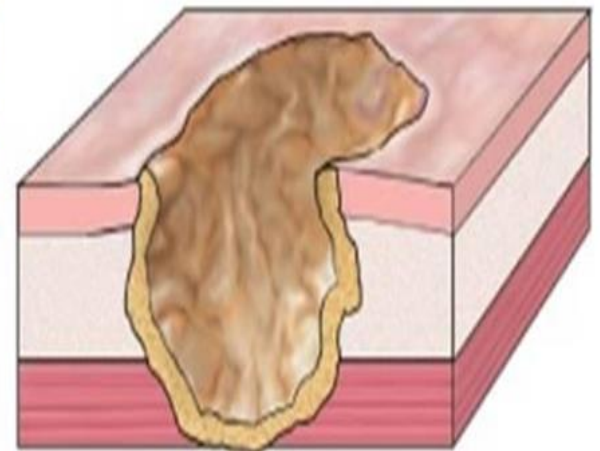
- Polypoidal (Exophytic)
- Infiltrative (Linitis plastica)
- Ulcerative (Excavated)



Exophytic



Linitis plastica



Excavated

Lauren classification:

According to their gross and histology appearance gastric adenocarcinoma include 2 types:-

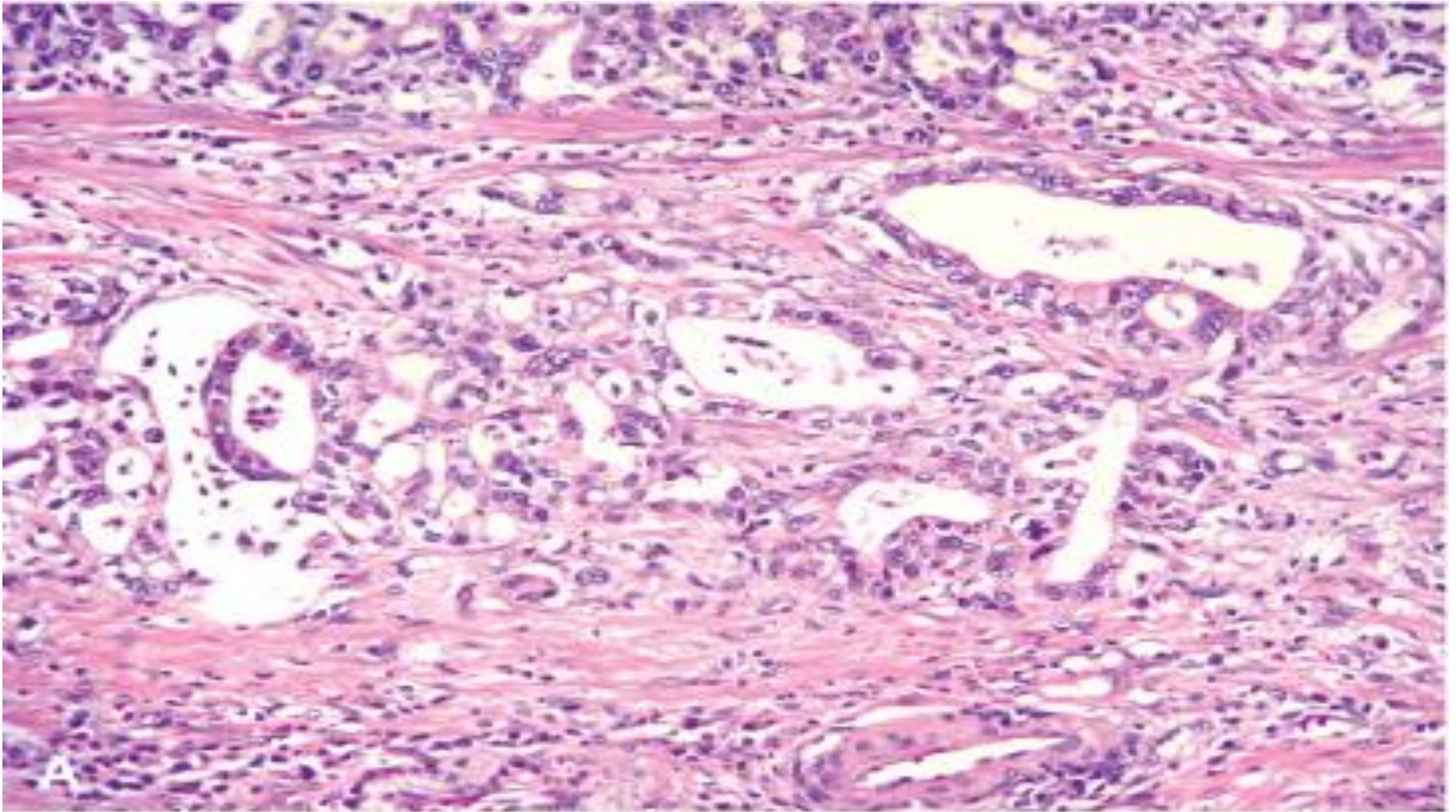
- Intestinal type
- Diffuse type

Intestinal-type:

- Composed of glandular structures similar to esophageal & colonic adenocarcinoma.
- Typically grows as an exophytic mass or an ulcerated tumor.
- Usually arise from precursor lesion & have risk factors.

Gastric cancer-intestinal type: Gross photograph showing an ill-defined, excavated central ulcer (Malignant gastric ulcer) surrounded by irregular, heaped-up borders, and dirty base.



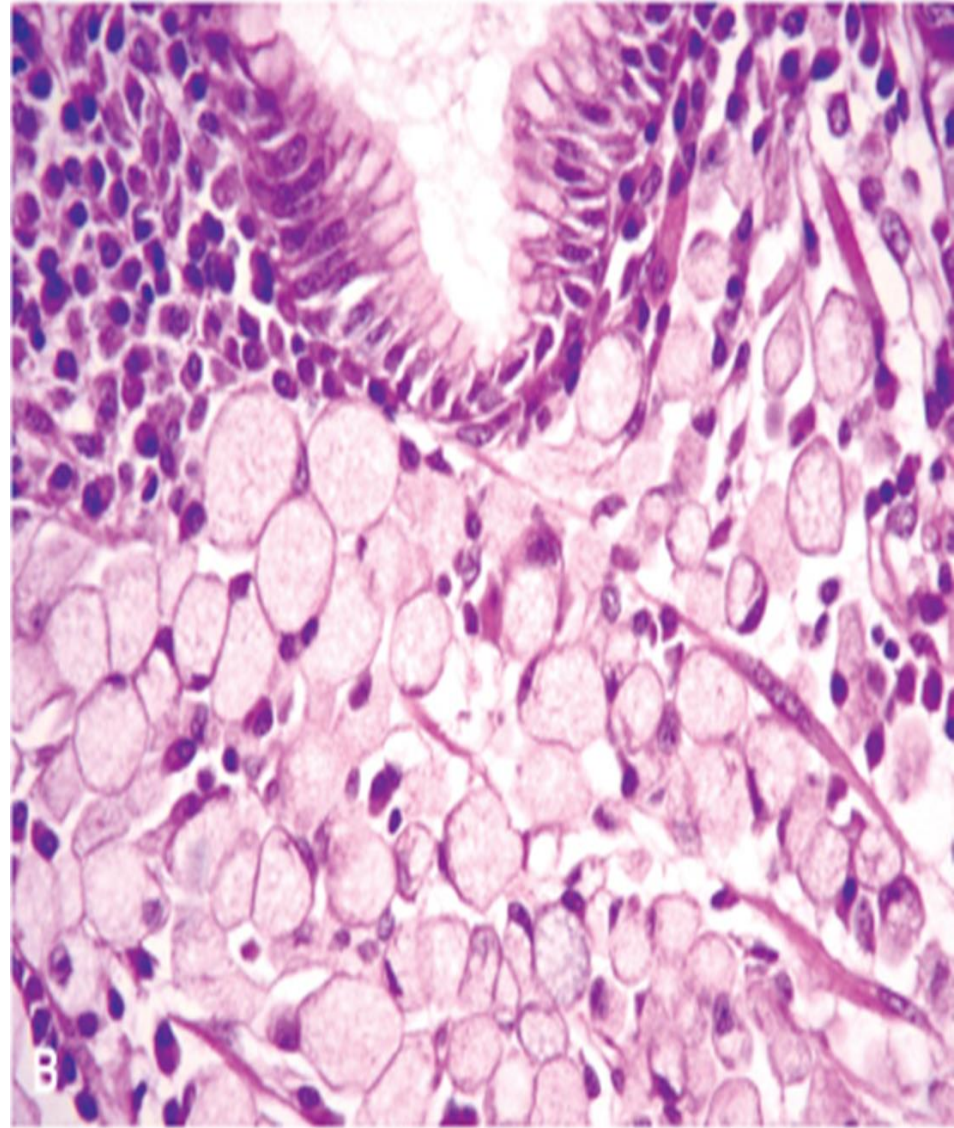


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Gastric carcinoma-Intestinal type: demonstrating gland formation by malignant cells, which are invading the muscular wall of the stomach.

Diffuse – type:

- Composed of signet ring cell morphology
- Display an infiltrative growth pattern.
- Often evoke a desmoplastic reaction that stiffens the gastric wall and may cause diffuse rugal flattening and imparts a “leather bottle” appearance termed **Linitis plastica**.
- No precursor lesion & do not have known risk factors.

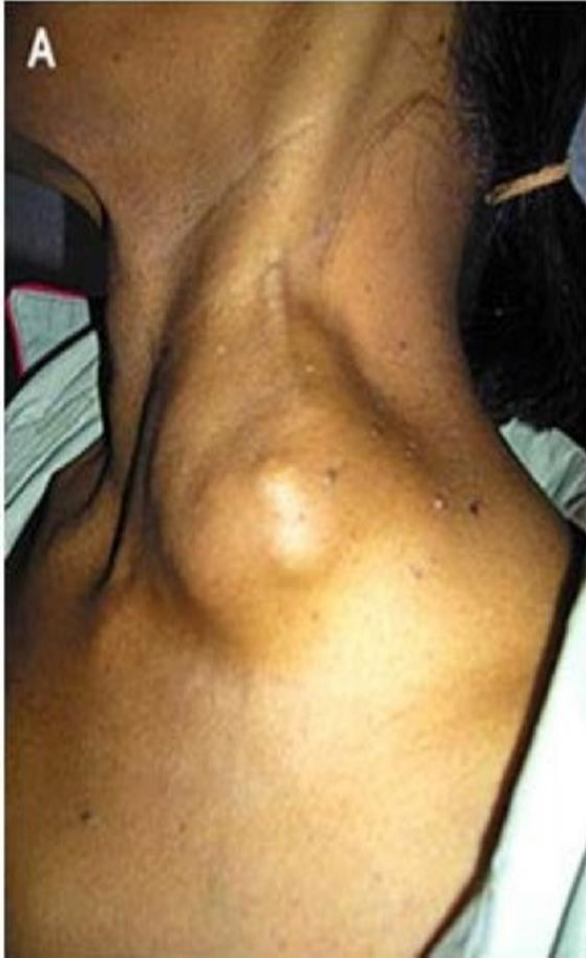


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Gastric cancer- diffuse type: gross (Linitis plastica); and microscope (malignant signet ring cell) appearance.

■ **Metastases:**

- Often detected at time of diagnosis.
- Sites commonly involved include:
 - Supraclavicular LN (Virchow node) / Troisier sign.
 - Periumbilical region (Sister Mary Joseph nodule),
 - Left axillary LN (Irish node),
 - Ovary (Krukenberg tumor),
 - Pouch of Douglas (Blumer shelf).



Troisier sign



Sister Mary Joseph's Nodule



Krukenberg tumor